Table of Basic Integrals [1]

(1) $$\int x^n\,dx = \frac{1}{n+1}x^{n+1},\ n \neq -1$$

(2) $$\int \frac{1}{x}\,dx = \ln|x|$$

(3) $$\int u\ dv = uv - \int v du$$

(4) $$\int e^x\,dx = e^x$$

(5) $$\int a^x\,dx = \frac{1}{\ln a}a^x$$

(6) $$\int \ln x\,dx = x\ln x - x$$

(7) $$\int \sin x\,dx = -\cos x$$

(8) $$\int \cos x\,dx = \sin x$$

(9) $$\int \tan x\,dx = \ln|\sec x|$$

(10) $$\int \sec x\,dx = \ln|\sec x + \tan x|$$

(11) $$\int \sec^2 x\,dx = \tan x$$

(12) $$\int \sec x \tan x\,dx = \sec x$$

(13) $$\int \frac{a}{a^2+x^2}\,dx = \tan^{-1}\frac{x}{a}$$

(14) $$\int \frac{a}{a^2-x^2}\,dx = \frac{1}{2}\ln\left|\frac{x+a}{x-a}\right|$$

(15) $$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\frac{x}{a}$$

(16) $$\int \frac{a}{x\sqrt{x^2-a^2}}\,dx = \sec^{-1}\frac{x}{a}$$

(17) $$\int \frac{1}{\sqrt{x^2-a^2}}\,dx = \cosh^{-1}\frac{x}{a} = \ln(x+\sqrt{x^2-a^2})$$

(18) $$\int \frac{1}{\sqrt{x^2+a^2}}\,dx = \sinh^{-1}\frac{x}{a} = \ln(x+\sqrt{x^2+a^2})$$

[1] Based on Material from `http://integral-table.com`, ©2011.

First Order ODEs

(19) **General Form of ODE:** $\frac{dy}{dx} = f(x,y)$

(20) **Initial Value Problem:** $y' = f(x,y),\ y(x_0) = y_0$

Linear Equations

(21) **General Form:** $y' + p(x)y = f(x)$

(22) **Integrating Factor:** $\mu(x) = e^{\int p(x)dx}$

(23) $$\implies \frac{d}{dx}(\mu(x)y) = \mu(x)f(x)$$

(24) **General Solution:** $y = \frac{1}{\mu(x)}\left(\int \mu(x)f(x)dx + C\right)$

Homeogeneous Equations

(25) **General Form:** $y' = f(y/x)$

(26) **Substitution:** $y = zx$

(27) $$\implies y' = z + xz'$$

The result is always separable in z:

(28) $$\frac{dz}{f(z)-z} = \frac{dx}{x}$$

Bernoulli Equations

(29) **General Form:** $y' + p(x)y = q(x)y^n$

(30) **Substitution:** $z = y^{1-n}$

The result is always linear in z:

(31) $$z' + (1-n)p(x)z = (1-n)q(x)$$

Exact Equations

(32) **General Form:** $M(x,y)dx + N(x,y)dy = 0$

(33) **Text for Exactness:** $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

(34) **Solution:** $\phi = 0$ where

(35) $$M = \frac{\partial \phi}{\partial x} \text{ and } N = \frac{\partial \phi}{\partial y}$$

Method for Solving Exact Equations:

1. Let $\phi = \int M(x,y)dx + h(y)$
2. Set $\frac{\partial \phi}{\partial y} = N(x,y)$
3. Simplify and solve for $h(y)$.
4. Substitute the result for $h(y)$ in the expression for ϕ from step 1 and then set $\phi = 0$. This is the solution.

Alternatively:

1. Let $\phi = \int N(x,y)dy + g(x)$
2. Set $\frac{\partial \phi}{\partial x} = M(x,y)$
3. Simplify and solve for $g(x)$.
4. Substitute the result for $g(x)$ in the expression for ϕ from step 1 and then set $\phi = 0$. This is the solution.

Integrating Factors

Case 1: If $P(x,y)$ depends only on x, where

(36) $$P(x,y) = \frac{M_y - N_x}{N} \implies \mu(y) = e^{\int P(x)dx}$$

then

(37) $$\mu(x)M(x,y)dx + \mu(x)N(x,y)dy = 0$$

is exact.

Case 2: If $Q(x,y)$ depends only on y, where

(38) $$Q(x,y) = \frac{N_x - M_y}{M} \implies \mu(y) = e^{\int Q(y)dy}$$

Then

(39) $$\mu(y)M(x,y)dx + \mu(y)N(x,y)dy = 0$$

is exact.

Second Order Linear ODEs

General Form of the Equation

(40) **General Form:** $a(t)y'' + b(t)y' + c(t)y = g(t)$

(41) **Homogeneous:** $a(t)y'' + b(t)y' + c(t) = 0$

(42) **Standard Form:** $y'' + p(t)y' + q(t)y = f(t)$

The **general solution** of (40) or (42) is

(43) $$y = C_1y_1(t) + C_2y_2(t) + y_p(t)$$

where $y_1(t)$ and $y_2(t)$ are linearly independent solutions of (41).

Linear Independence and The Wronskian

Two functions $f(x)$ and $g(x)$ are **linearly dependent** if there exist numbers a and b, not both zero, such that $af(x) + bg(x) = 0$ for all x. If no such numbers exist then they are **linearly independent**.
If y_1 and y_2 are two solutions of (41) then

(44) **Wronskian:** $W(t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$

(45) **Abel's Formula:** $W(t) = Ce^{-\int p(t)dt}$

and the following are all equivalent:

1. $\{y_1, y_2\}$ are linearly independent.
2. $\{y_1, y_2\}$ are a fundamental set of solutions.
3. $W(y_1, y_2)(t_0) \neq 0$ at some point t_0.
4. $W(y_1, y_2)(t) \neq 0$ for all t.

Initial Value Problem

(46) $$\begin{cases} y'' + p(t)y' + q(t)y = 0 \\ y(t_0) = y_0 \\ y'(t_0) = y_1 \end{cases}$$

Linear Equation: Constant Coefficients

(47) **Homogeneous:** $ay'' + by' + cy = 0$

(48) **Non-homogeneous:** $ay'' + by' + cy = g(t)$

(49) **Characteristic Equation:** $ar^2 + br + c = 0$

(50) **Quadratic Roots:** $r = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The solution of (47) is given by:

(51)
Real Roots$(r_1 \neq r_2)$: $y_H = C_1e^{r_1t} + C_2e^{r_2t}$

(52) **Repeated**$(r_1 = r_2)$: $y_H = (C_1 + C_2t)e^{r_1t}$

(53)
Complex$(r = \alpha \pm i\beta)$: $y_H = e^{\alpha t}(C_1 \cos\beta t + C_2 \sin\beta t)$

The solution of (48) is $y = y_P + h_H$ where y_h is given by (51) through (53) and y_P is found by **undetermined coefficients** or **reduction of order**.

Heuristics for Undetermined Coefficients (Trial and Error)

If $f(t) =$	then guess that $y_P =$
ke^{at}	Ce^{at}
kt^n	$K_nt^n + K_{n-1}t^{n-1} + \cdots + K_1t + K_0$
$\sin\omega t$ or $\cos\omega t$	$K\cos\omega t + M\sin\omega t$
t^ne^{at}	$t^s(A_0 + A_1t + \cdots + A_nt^n)e^{at}$
$ke^{\alpha t}\cos\omega t$ or $ke^{at}\sin\omega t$	$e^{\alpha t}(K\cos\omega t + M\sin\omega t)$

Method of Reduction of Order

When solving (41), given y_1, then y_2 can be found by solving

(54) $$y_1y_2' - y_1'y_2 = Ce^{-\int p(t)dt}$$

The solution is given by

(55) $$y_2 = y_1 \int \frac{e^{-\int p(x)dx}dx}{y_1(x)^2}$$

Method of Variation of Parameters

If $y_1(t)$ and $y_2(t)$ are a fundamental set of solutions to (41) then a particular solution to (42) is

(56) $$y_P(t) = -y_1(t)\int \frac{y_2(t)f(t)}{W(t)}dt + y_2(t)\int \frac{y_1(t)f(t)}{W(t)}dt$$

Cauchy-Euler Equation

(57) **ODE:** $ax^2y'' + bxy' + cy = 0$

(58) **Auxilliary Equation:** $ar(r-1) + br + c = 0$

The solutions of (57) depend on the roots of (58):

(59) **Real Roots:** $y = C_1x^{r_1} + C_2x^{r^2}$

(60)
Repeated Root: $y = C_1x^r + C_2x^r \ln x$

(61) **Complex:** $y = x^\alpha[C_1\cos(\beta\ln x) + C_2\sin(\beta\ln x)]$

Series Solutions

(62) $$(x - x_0)^2y'' + (x - x_0)p(x)y' + q(x)y = 0$$

If x_0 is a **regular point** of (62) then

(63) $$y_1(t) = (x - x_0)^n \sum_{k=0}^{\infty} a_k(x - x_k)^k$$

At a **Regular Singular Point** x_0:

(64) **Indicial Equation:** $r^2 + (p(0) - 1)r + q(0) = 0$

(65) **First Solution:** $y_1 = (x - x_0)^{r_1} \sum_{k=0}^{\infty} a_k(x - x_k)^k$

Where r_1 is the larger real root if both roots of (64) are real or either root if the solutions are complex.

Laplace Transforms

$f(t)$	$\mathscr{L}[f(t)] = F(s)$	
1	$\dfrac{1}{s}$	(1)
$e^{at}f(t)$	$F(s-a)$	(2)
$\mathcal{U}(t-a)$	$\dfrac{e^{-as}}{s}$	(3)
$f(t-a)\mathcal{U}(t-a)$	$e^{-as}F(s)$	(4)
$f'(t)$	$sF(s) - f(0)$	(5)
$f^n(t)$	$s^nF(s) - s^{(n-1)}f(0) - \cdots - f^{(n-1)}(0)$	(6)

$f(t)$	$\mathscr{L}[f(t)] = F(s)$	
t^n $(n = 0, 1, 2, \dots)$	$\dfrac{n!}{s^{n+1}}$	(7)
$\sin kt$	$\dfrac{k}{s^2 + k^2}$	(8)
$\cos kt$	$\dfrac{s}{s^2 + k^2}$	(9)
e^{at}	$\dfrac{1}{s - a}$	(10)
$\sinh kt$	$\dfrac{k}{s^2 - k^2}$	(11)
$\cosh kt$	$\dfrac{s}{s^2 - k^2}$	(12)